

Lilian is the project manager of Jokete Construction Company. The Company is bidding on a affordable housing project to construct 1000 units in Ngala. They have identified the following activities along with their predecessor restrictions, expected times and worker requirements.

Activity	Predecessors	Duration Weeks	Crew Size Workers
A	-	4	4
B	-	7	2
C	A	3	2
D	A	3	4
E	B	2	6
F	B	2	3
G	D, E	2	3
H	F, G	3	4

Lilian has agreed with the client that the project should be completed in the shortest duration.

Required:

- (i) Draw a network for the project.
- (ii) Lilian will assign a fixed number of workers to the project for its entire duration and so she would like to ensure that the minimum number of workers is assigned and that the project will be completed in 14 weeks. Draw a schedule showing how the project will be completed in 14 weeks.
- (iii) Determine the critical path and the shortest project duration.
- (iv) Comment on the schedule you have drawn in (iii) above.